Exercise 1

In the lecture we convinced ourselves that all maps are well-defined. As an inclusion, the first map is clearly injective.

Exactness at $\mathcal{O}_k^\times$ is clear, since $\rho(u) = 1 \in (\mathcal{O}_k/\mathfrak{m})^\times$ by definition means exactly $u \equiv 1 \mod {}^\times\mathfrak{m}$.

Exactness at $(\mathcal{O}_k/\mathfrak{m})^\times$: For $u \in \mathcal{O}_k^\times$ we have $(u) = (1)$, so $(u)\mathcal{P}_k(\mathfrak{m}) = \mathcal{P}_k(\mathfrak{m})$. Conversely, let $\alpha \in k^\times$, $(\alpha, \mathfrak{m}_0) = 1$ and suppose $(\alpha)\mathcal{P}_k(\mathfrak{m}) = \mathcal{P}_k(\mathfrak{m})$. That is, there is some $\varepsilon \in \mathcal{O}_k^\times$ s.t. $\varepsilon\alpha \equiv 1 \bmod {}^\times\mathfrak{m}$, and $\rho(\alpha) = \rho(\varepsilon^{-1}) \in \rho(\mathcal{O}_k^\times)$.

Exactness at $\mathrm{cl}_k(\mathfrak{m})$ is clear, since both kernel and image consist exactly of the classes of principal ideals coprime to $\mathfrak{m}$.

Finally, we show more generally, that if $\mathfrak{m} \mid \mathfrak{n}$, then the induced map $\mathrm{cl}_k(\mathfrak{n}) \to \mathrm{cl}_k(\mathfrak{m})$ is surjective: Let $\mathfrak{a}\mathcal{P}_k(\mathfrak{m}) \in \mathrm{cl}_k(\mathfrak{m})$, for some ideal $\mathfrak{a} \in I_k(\mathfrak{m})$. By the approximation theorem, we find $\alpha \in k$ with $\alpha \equiv 1 \bmod {}^\times\mathfrak{m}$ and $\nu_\mathfrak{p}(\alpha) = -\nu_\mathfrak{p}(\mathfrak{a})$ for all prime ideals $\mathfrak{p} \mid (\mathfrak{n}, \mathfrak{a})$. Then $(\alpha\mathfrak{a}, \mathfrak{n}) = 1$, and

$$\alpha\mathfrak{a}\mathcal{P}_k(\mathfrak{n}) \mapsto \alpha\mathfrak{a}\mathcal{P}_k(\mathfrak{m}) = \mathfrak{a}\mathcal{P}_k(\mathfrak{m}).$$

Exercise 2

If $0 \to A \to B \to C \to 0$ is a short exact sequence of groups, then B is partitioned into $|C|$ cosets each of size $|A|$, hence $|B| = |A| \cdot |C|$. If $0 \to A \to B \to C \to D \to 0$ is exact, let $K = \ker(C \to D) = \mathrm{im}(A \to B)$. Then $0 \to A \to B \to K \to 0$ and $0 \to K \to C \to D \to 0$ are exact, and therefore

$$|A| \cdot |C| = |A| \cdot |K| \cdot |D| = |B| \cdot |D|.$$

(More generally, given any finite exact sequence, continuing in this way yields an equation of "alternating" cardinalities.)

Applying this to the exact sequence $0 \to \mathcal{O}_k^\times/(\mathcal{O}_k^\times \cap k_\mathfrak{m}^\times) \to \ldots \to 0$ yields

$$[\mathcal{O}_k^\times : \mathcal{O}_k^\times \cap k_\mathfrak{m}^\times] \cdot |\,\mathrm{cl}_k(\mathfrak{m})| = |\,\mathrm{cl}_k\,| \cdot |(\mathcal{O}_k/\mathfrak{m})^\times|.$$

In particular, since the right hand side is finite, and groups cannot be empty, both cardinalities on the left hand-side are finite as well, and dividing by the first term yields the result.

Exercise 3

Let $k = \mathbb{Q}(\sqrt{d})$ and let $\tau_1 = \mathrm{id}, \tau_2 : \bar{\cdot}$ be the two embeddings associated to $\infty_{1/2}$. By Dirichlet, $\mathcal{O}_K^\times = \{\pm 1\} \times \varepsilon^\mathbb{Z}$, where wlog $\varepsilon > 1$. Clearly $-1 \not\equiv 1 \bmod {}^\times\mathfrak{m}$. From $\mathrm{N}_{k/\mathbb{Q}}(\varepsilon) = \varepsilon\bar{\varepsilon}$ we see $\mathrm{sgn}(\bar{\varepsilon}) = \mathrm{N}_{k/\mathbb{Q}}(\varepsilon)$. Hence if $\mathrm{N}_{k/Q}(\varepsilon) = 1$, then $\mathcal{O}_k^\times \cap k_\mathfrak{m}^\times = \varepsilon^\mathbb{Z}$ and the index in $\mathcal{O}_k^\times$ is 2. Otherwise, $\mathcal{O}_k^\times \cap k_\mathfrak{m}^\times = \varepsilon^{2\mathbb{Z}}$ and the index in $\mathcal{O}_k^\times$ is 4.

Further, since $\mathfrak{m}_0 = (1)$ we have $|(\mathcal{O}_k/\mathfrak{m})^\times| = |\{\pm 1\}^2| = 4$. The result now follows from exercise 2.

Exercise 1

We have

$$\frac{(\mathcal{O}_K/\mathfrak{f})^\times}{U} \cong \ker(\mathrm{cl}_k(\mathfrak{f}) \to \mathrm{cl}_k) \cong \ker(\mathrm{Gal}(k(\mathfrak{f})/k) \to \mathrm{Gal}(k(1)/k)) \cong \mathrm{Gal}(k(\mathfrak{f})/k(1)),$$

where the first isomorphism follows from last week's sheet, the central one is induced by the Artin map, and the last is Galois theory.

Exercise 2

Let $\mathfrak{m} := q\mathbb{Z}\infty$. Then we know $\mathbb{Q}(\mathfrak{m}) = \mathbb{Q}(\zeta_q)$ and hence the Artin map induces an isomorphism

$$\mathrm{cl}_\mathbb{Q}(\mathfrak{m}) \cong \mathrm{Gal}(\mathbb{Q}(\zeta_q)/\mathbb{Q}) \cong (\mathbb{Z}/q\mathbb{Z})^\times \cong \mathbb{Z}/(q-1)\mathbb{Z}.$$

Let $H \subseteq \mathrm{cl}_Q(\mathfrak{m})$ be the unique subgroup of index 2. Then the existence theorem yields a unique abelian subfield $\mathbb{Q}(\zeta_q)/K/\mathbb{Q}$ with $\mathrm{Gal}(K/\mathbb{Q}) \cong \mathrm{cl}_\mathbb{Q}(\mathfrak{m})/H$, and ν ramifies in K only if $\nu \mid \mathfrak{m}$. Now the first condition implies that $K/\mathbb{Q}$ is quadratic, say $K = \mathbb{Q}(\sqrt{d})$ for some squarefree $d \in \mathbb{Z}$. Let p be a prime divisor of d. Then p ramifies in K, hence $p \mid q$. Hence either $d = -1$, or $d = \pm q$. Now 2 isn't allowed to ramify, so $d = -1$ is out, and we need the sign so that $\pm q \equiv 1 \bmod 4$. Hence if $q \equiv 1 \bmod 4$, we have $\mathbb{Q}(\sqrt{q}) \subseteq \mathbb{Q}(\zeta_q)$, and if $q \equiv 3 \bmod 4$, then $\mathbb{Q}(\sqrt{-q}) \subseteq \mathbb{Q}(\zeta_q)$, hence

$$\mathbb{Q}(\sqrt{q}) \subseteq \mathbb{Q}(\sqrt{-q}, \sqrt{-1}) = \mathbb{Q}(\sqrt{-q})\mathbb{Q}(\zeta_4) \subseteq \mathbb{Q}(\zeta_q)\mathbb{Q}(\zeta_4) = \mathbb{Q}(\zeta_{4q}).$$

In both cases this implies the claim, since $\mathbb{Q}(\sqrt{q})$ is totally real.

Exercise 3

Let $m := [k : \mathbb{Q}_p]$. Then we showed in ANT, that

$$k^\times \cong \mathbb{Z} \oplus \mathbb{Z}/(q-1)\mathbb{Z} \oplus \mathbb{Z}/p^a\mathbb{Z} \oplus \mathbb{Z}_p^m.$$

for a p-power q and non-negative integer a. Hence

$$k^\times/(k^\times)^n \cong \mathbb{Z}/n\mathbb{Z} \oplus \mathbb{Z}/(n, q-1)\mathbb{Z} \oplus \mathbb{Z}/(n, p^a)\mathbb{Z} \oplus (\mathbb{Z}_p/n\mathbb{Z}_p)^m.$$

The first three summands are clearly finite, hence it remains to consider $\mathbb{Z}_p/n\mathbb{Z}_p$. Write $n = p^v b$ with $v, b \geq 0$, $p \nmid b$. Then b is invertible in $\mathbb{Z}_p$, hence $\mathbb{Z}_p/n\mathbb{Z}_p \cong \mathbb{Z}_p/p^v\mathbb{Z}_p$, which is isomorphic to $\mathbb{Z}/p^v\mathbb{Z}$ by ANT. Putting everything together, we see that $(k^\times)^n$ has finite index in $k^\times$.

To see that $(k^\times)^n$ is open, it suffices to find an $N \in \mathbb{N}$ with $1 + \mathfrak{p}_k^N \subseteq (k^\times)^n$, then $(k^\times)^n = \bigcup_{\alpha \in (k^\times)^n} \alpha(1 + \mathfrak{p}_k^N)$ is a union of open sets, hence open. To do this, let $s > \frac{e}{p-1}$ and $N = s + v_k(n)$. Then

$$1 + \mathfrak{p}_k^N = \exp(n\mathfrak{p}_k^s) = \exp(\mathfrak{p}_k^s)^n = (1 + \mathfrak{p}_k^s)^n \subseteq (k^\times)^n.$$

Exercise 4

Let K/k be finite abelian of exponent p. By local class field theory, for $a \in k^\times$ we have $(a^p, K/k) = (a, K/k)^p = \mathrm{id} \in \mathrm{Gal}(K/k)$. Hence $(k^\times)^p \subseteq \ker((\_, K/k))$. Conversely, if $(k^\times)^p \subseteq \ker((\_, K/k))$, then since the Artin map is surjective, $\mathrm{Gal}(K/k)$ has exponent (dividing) p. Hence the finite abelian extensions of exponent p correspond exactly to the subgroups $(k^\times)^p \subseteq H \subsetneq k^\times$ (they are automatically open).

By the same calculation as in exercise 3, for $p \neq 2$ we have $k^\times/(k^\times)^p \cong \mathbb{Z}/p\mathbb{Z} \oplus \mathbb{Z}/p\mathbb{Z}$. This has $\frac{p^2-1}{p-1} = p+1$ subgroups of order p, and the trivial subgroup, so there are $p+2$ finite abelian extensions of $\mathbb{Q}_p$ of exponent p.

For $p = 2$, the same method yields $k^\times/(k^\times)^2 \cong (\mathbb{Z}/2\mathbb{Z})^3$, which has 7 subgroups of order 2 , and $\frac{1}{3}\binom{7}{2} = 7$ subgroups of order 4, so there are $1+7+7 = 15$ finite abelian extensions of $\mathbb{Q}_2$ of exponent 2.